

CELLULAR AUTOMATA IN CYTOSKELETAL LATTICES

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Cellular automata (CA) activities could mediate biological regulation and information processing via nonlinear electrodynamic effects in cytoskeletal lattice arrays. Frohlich coherent oscillations and other nonlinear mechanisms may effect discrete 10^{-10} to 10^{-11} s interval events which result in dynamic patterns in biolattices such as cylindrical protein polymers: microtubules (MT). Structural geometry and electrostatic forces of MT subunit dipole oscillations suggest neighbor rules among the hexagonally packed protein subunits. Computer simulations using these suggested rules and MT structural geometry demonstrate CA activities including dynamical and stable self-organizing patterns, oscillators, and traveling “gliders”. CA activities in MT and other cytoskeletal lattices may have important biological regulatory functions.

1. Introduction

Cellular automata (CA) are dynamical systems which can generate and process patterns and information via lattice neighbor interactions [1–3]. Various biomolecular systems have lattice structures and participate in organized dynamical functions. These structures include membrane lipids and proteins, extracellular glycoproteins, DNA, and the cytoskeleton: microtrabecular lattice, protein filaments and microtubules (MT) [4–7] (fig. 1). Coherent dipole oscillations resulting in long-range cooperativity, as well as other nonlinear electrodynamic effects have been theorized to regulate activities in these structures [8–9]. Coherent oscillations among biolattice grid subunits could mediate information processing via CA-like activities. This paper considers the possibilities and potential significance of CA-like activities in a particular biolattice: cytoskeletal MT.

2. Coherent excitations in biological systems

Frohlich has theorized that non-linear dipole excitations among subunits of biomolecular arrays

will result in coherent oscillations (10^{-10} to 10^{-11} s fluctuations), lower frequency metastable states, and long range order [8–9]. Frohlich also theorized that protein conformational states are coupled to electron dipole oscillations within protein hydrophobic regions [8]. Conformational coupling to coherent electronic excitation by individual sub-

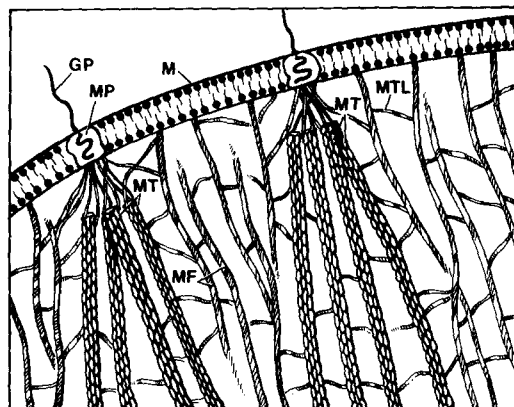


Fig. 1. Schematic representation of cytoskeletal lattice structures. M: cell membrane, MP: membrane protein (ion channel, receptor etc.), GP: glycoproteins which extend into extracellular space, MT: microtubules, MF: microfilaments including actin, intermediate, and neurofilaments, MTL: microtrabecular lattice. MT, MF, and MTL comprise the cytoskeletal lattice which interconnects and organizes cellular structures and activities [6].

units in biomolecular arrays would apparently depend on various (genetic and environmental) biochemical factors as well as neighbor interactions. Neighbor interactions in a protein polymer can be mediated by factors which include electrostatic dipole interactions, water and calcium ion binding, conformational changes, binding of various substances including GTP and associated proteins, intrinsic subunit factors, or solitons [10, 11]. Such influences could alter the oscillatory phase of particular subunits in a coherent biomolecular array, resulting in dynamic patterns and CA-like activities.

Dynamic and stable patterns of excitation and conformational states in biomolecular structures could have direct, profound effects on regulation and organization of cellular processes. Patterns of resonant excitation and conformational states in various structures could mediate protein binding sites, ion fluxes, transport of biomolecules, growth, organization of cellular activities, and information processing. CA are thus theoretically possible and of potential significance in biological systems.

3. Cytoskeletal lattices

Cytoskeletal lattices including protein polymers, microtubules (MT), actin, intermediate filaments, and the microtrabecular lattice interconnect and organize nearly all cell components and dynamic activities (fig. 1). Intracellular movement and coordination, growth, mitosis, transport and turnover of membrane proteins are accomplished by cytoskeletal lattices and attached proteins. MT are cylindrical polymers which comprise cilia, mitotic spindles, other organelles, and are the major cellular architectural elements [6, 12]. MT are intimately involved in dynamic biological activities, but mechanisms of "real time" regulation and control by MT or other structures are unknown [13]. The particular structure and geometry of MT will be considered as potential CA lattices.

Most MT are assemblies of 13 longitudinal protofilaments which are each a series of polar,

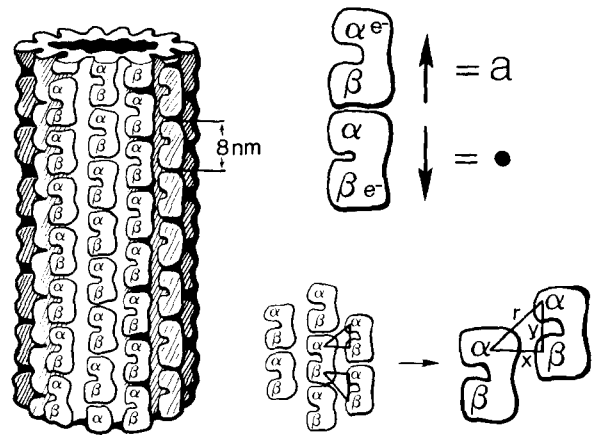


Fig. 2. Left: microtubule (MT) structure from X-ray diffraction crystallography [15]. Right top: MT tubulin dimer subunits comprised of α and β monomers. Electron occupancy of either monomer may correlate with coherent dipole oscillations and are represented [figs. 3–6] as "a" for α monomer electron occupancy and "." for β monomer electron occupancy. Center bottom and right: hexagonal relations lead to neighbor rules based on lattice distances. For neighbors shown (lower right), $x = 5$ nm, $y = 4$ nm, $r = 6.4$ nm.

tubulin dimers (fig. 2). A dimer consists of two slightly different 55,000 molecular weight monomers (α and β tubulin) which are each 4 nm in diameter. Hexagonal packing of the dimers results in 5 and 8 start helical patterns [14–16]. MT outer diameters are 25–30 nm and inner diameters are 14–15 nm [17]. MT outer surfaces are surrounded by a "clear zone" of several nm which apparently represents electronegative charges on tubulin [18]. Functional MT lengths may range from hundreds of nm to micrometers and meters in some mammalian neurons [19, 20]. Contractile or enzymatic proteins may be attached at specific dimer sites resulting in various helical patterns [16]. Various proteins are transported by MT at rates from one to 400 millimeters/day [21, 22]. Also binding reversibly at specific dimer sites are GTP molecules, hydrolysis of which can yield phosphate bond energies capable of oscillations or soliton formation [23]. Thus programming modes by genetic or environmental factors can introduce solitons or oscillatory phase shifts at specific lattice sites.

Table I

Neighbor position (fig. 2)	Central dimer α		Central dimer β	
	Neighbor α	Neighbor β	Neighbor α	Neighbor β
North	+ 0.063	+ 0.254	+ 0.028	+ 0.063
Northeast	+ 0.063	+ 0.007	+ 0.040	+ 0.063
Southeast	- 0.061	- 0.037	- 0.007	- 0.061
South	- 0.063	- 0.028	- 0.254	- 0.063
Southwest	- 0.063	- 0.040	- 0.007	- 0.063
Northwest	+ 0.061	+ 0.007	+ 0.037	+ 0.061

Configuration of tubulin dimer subunits may favor dipole oscillations between α and β states (fig. 2). In the context of Frohlich coherent excitation, various factors could modulate oscillatory phase of individual lattice subunits. (See section on

Coherent excitations in biological systems.) The most predictable and consistent of these factors, electrostatic forces due to MT subunit neighbor dipoles, will be considered as the rules for CA activity in the structural geometry of MT lattices.

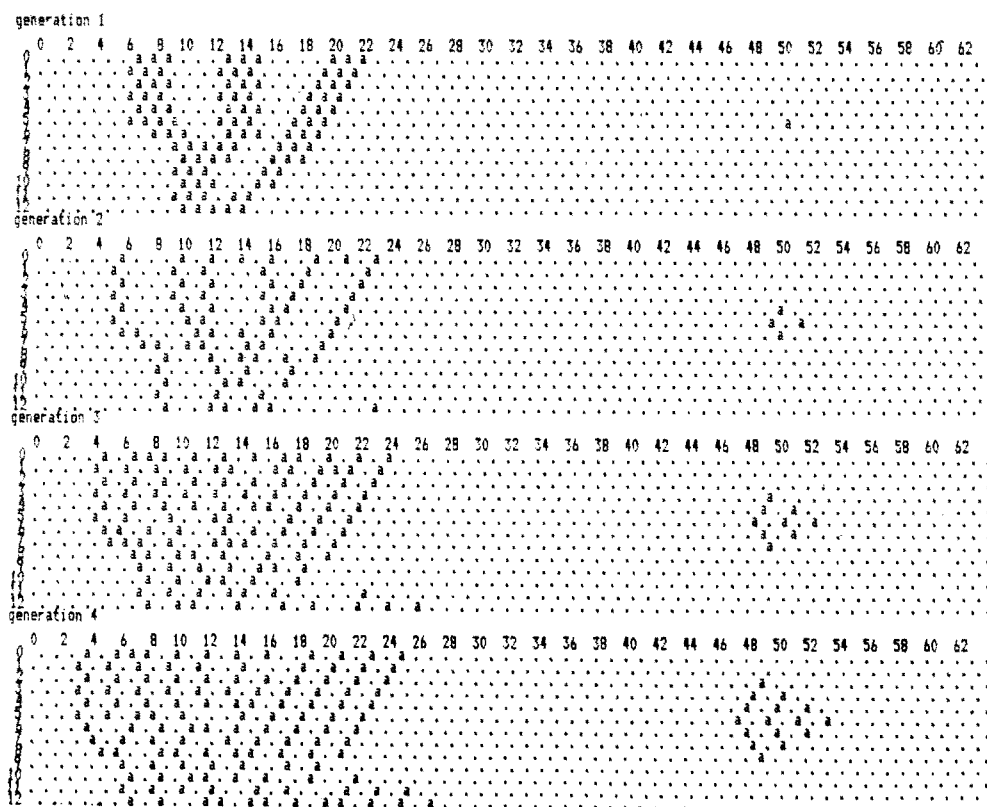


Fig. 3. Unfolded MT lattice with 13 protofilaments (0–12) on vertical axis, and “north” neighbors and dimer orientation on horizontal axis. An organized (arbitrary) pattern of α states starts on the left side and a single α state originates on the right. In generations 1–4 (generation numbers are on top of accompanying lattice) both the pattern and single α state develop and grow according to rules described in text.

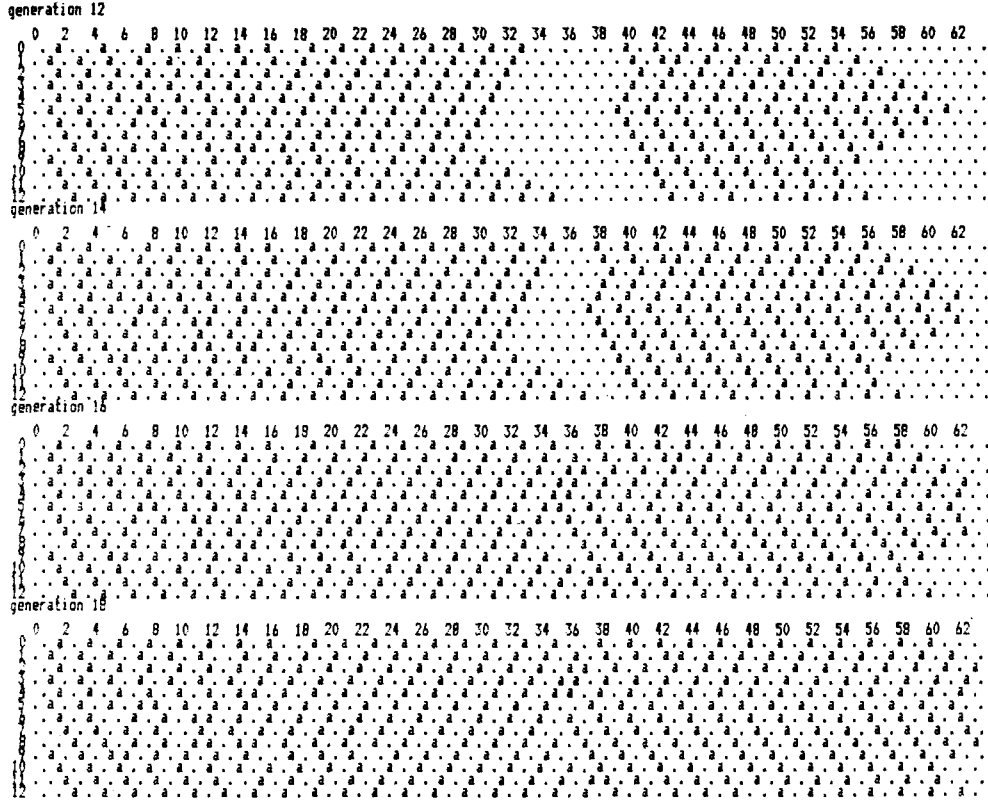


Fig. 4. Generations 12, 14, 16, and 18 of the same sequence as fig. 3. The two patterns have evolved and begin to interact.

4. CA rules for MT lattices

Frohlich's theory predicts 10^{-10} to 10^{-11} s coherent dipole oscillations of MT subunit dimers. If dipole polarities may exist in either the α and β orientation (fig. 2) then the state of each dimer at each oscillation may depend largely on neighbor dipole electrostatic forces.

Relative contributions of neighbor dipoles will consist of the y -component of the electrostatic force (along the axis of the dimers and protofilaments). The state of a dimer in the subsequent 10^{-10} s generation may thus be expressed:

$$\text{state} = \alpha, \quad \text{if } \sum_{i=1}^n f(y) > 0,$$

where n is the number of neighbors ($n = 7$);

$$\text{state} = \beta, \quad \text{if } \sum_{i=1}^n f(y) < 0,$$

$f(y)$ is the force from the " i th" neighbor in the y direction,

$$f(y) \propto \frac{\sin \theta}{r^2}, \quad \sin \theta = \frac{y}{r}; \quad f(y) \propto \frac{y}{r^3}.$$

x , y , and r are distances and θ the angle between neighboring monomers (fig. 2).

Thus states may be determined by $\sum_{i=1}^n y/r^3$ and relative values calculated from known MT structural geometry; see table I.

5. Computer simulation of CA patterns in MT lattices

Using the described rules, the following computer simulations were done. The cylindrical MT structures are represented as unfolded, horizontal ("North" neighbors are now "East") planes. α states are shown as "a", and β states by dots (figs. 2–6). The number of each successive generation indicating an oscillation period of 10^{-10} to 10^{-11} s is shown at the top of each lattice. In fig. 3, an organized pattern of "a" states starts on the left side, and a single, isolated "a" state starts on the right side. In generations 1–4 both the pattern and single state develop. Fig. 4 demonstrates gener-

ations 12, 14, 16 and 18 in which the 2 patterns begin to interact. In figs. 5 and 6, the β dots have been connected to enhance contrast. Fig. 5 demonstrates generations 22, 28, 34, and 40 from the same sequence. Even numbered stable patterns have evolved. The right side reveals a traveling pattern moving at maximum velocity (8 nm/nanosecond if based on 10^{-11} s oscillations) and leaving an altered wake. In fig. 6 generations 44, 46, 47, and 48 demonstrate the traveling pattern, which may be likened to a "glider" [3], Frohlich metastable state, or soliton reaching the end of the display. A single odd numbered generation (47) is a stable oscillator which includes a noticeable crossing pattern. Thus CA rules based

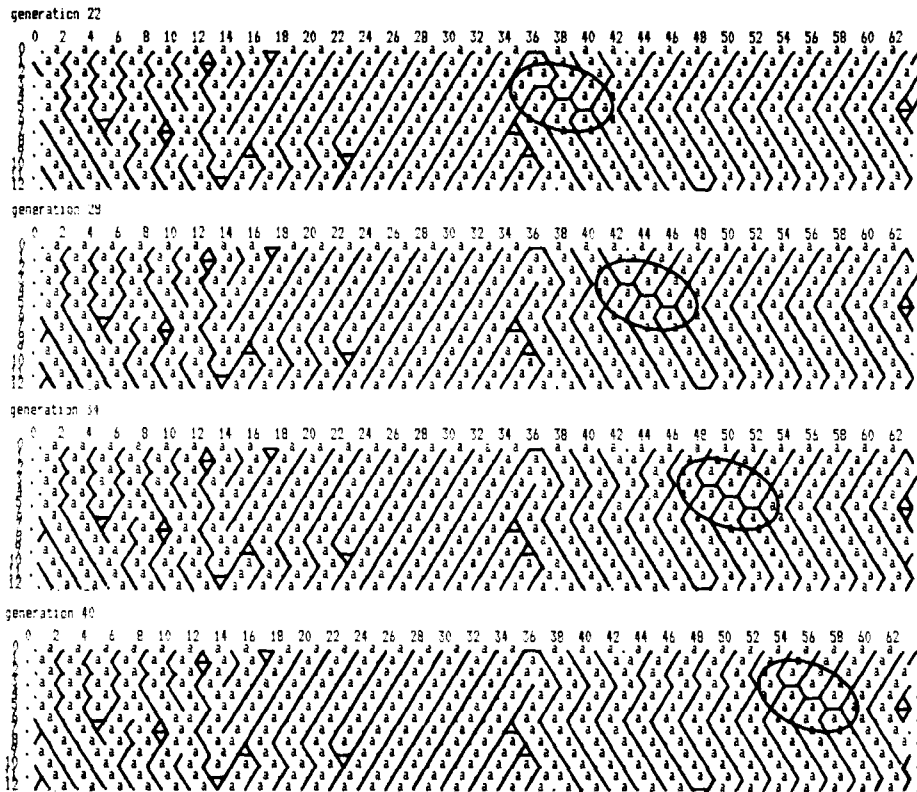


Fig. 5. Generations 22, 28, 34, and 40 from the same sequence. In figs. 5 and 6, β states represented by dots have been connected by lines to enhance contrast. Stable patterns have evolved (during even numbered generations) except for a traveling pattern which may be likened to a "glider", and the "glider's" altered "wake".

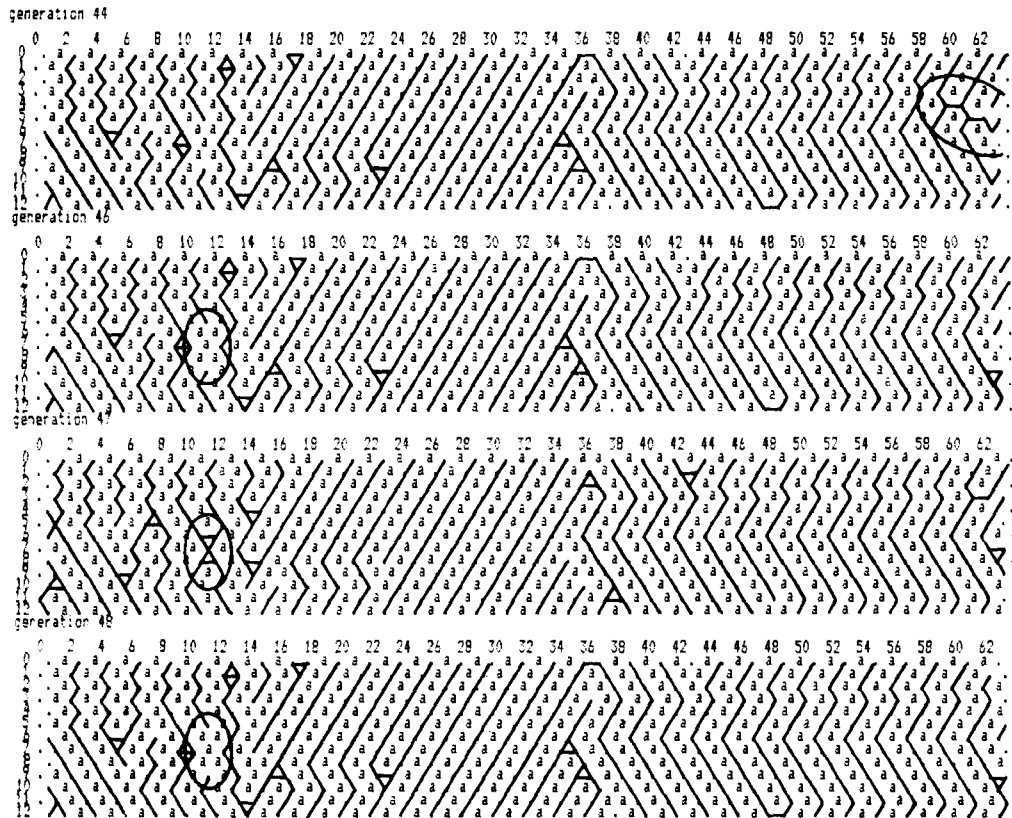


Fig. 6. Generations 44, 46, 47, and 48 demonstrate the “glider” leaving the display. Odd numbered generation 47 reveals a stable oscillator patterns which includes a noticeable crossing pattern.

on electrostatic dipole interactions in cytoskeletal geometries can result in stable patterns, oscillators, and traveling patterns (“gliders”).

6. Conclusion

Stable and moving patterns of MT subunit oscillation phase could function as specific sites of protein binding and transport, orchestration of biomolecular activities, as well as stored information or memory. Interaction and interference of these patterns including “gliders” could process information as in computers. Although the CA rules and patterns described are somewhat arbitrary, they demonstrate that CA-like activities in

cytoskeletal lattices are thus feasible and of potentially great importance.

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